CAPSTONE PROJECT 32

AI Research Paper Explainer for Undergraduates

A practical project report for Information Technology students

Project question	Can AI make complex research papers understandable for undergraduate students?
Field	Artificial Intelligence in Higher Education
Tools proposed	NotebookLM, ChatGPT, Gemini, Canva, and Gamma/PowerPoint
Prepared on	17 October 2026

Important note: This report provides a ready-to-edit project structure and literature-based summaries. Replace the blank student details, run the prompts yourself, verify all AI outputs against the full papers, and collect real classmate feedback before final submission. The report does not claim that user testing has already been completed.

Abstract

Research papers often use specialized terminology, long methods sections, statistics, and unfamiliar concepts. This project explores whether AI tools can help undergraduate students understand research papers more easily while preserving accuracy. Three open-access review papers about AI in higher education are selected. Their aims, methods, findings, and limitations are rewritten in beginner-friendly language. AI-generated summaries are then checked against the original papers, converted into visual summaries, and used to prepare a short presentation. A verification log and a student-feedback form are included so that the process can be evaluated transparently. The key principle is that AI is a learning assistant, not a replacement for reading the original research or checking evidence.

Keywords: artificial intelligence, research paper, undergraduate learning, summarization, verification, responsible AI

1. Introduction

Academic research helps students learn about new ideas, methods, and evidence. However, students may find it difficult to identify a paper's research question, understand the methodology, interpret results, or recognize limitations. AI tools can summarize and explain technical content, create questions, and reorganize information. At the same time, they may invent details, misstate statistics, omit caveats, or present uncertain findings too confidently. Therefore, this project combines AI-assisted explanation with human verification.

2. Problem statement

Undergraduate students may struggle to understand research papers because of technical vocabulary, unfamiliar research methods, and dense academic writing. The project investigates a workflow that uses AI to make papers easier to understand while checking every important claim against the source.

3. Aim and objectives

- To simplify three open-access research papers for undergraduate IT students.
- To generate summaries, beginner-level explanations, and frequently asked questions using AI tools.
- To create one-page visual summaries and a five-slide presentation.
- To identify and document inaccurate, unsupported, or incomplete AI statements.
- To collect student feedback and assess whether the explanations are understandable.
- To document responsible AI use, limitations, prompts, and reproducible steps.

4. Scope

This is a small educational project, not a new AI model and not a clinical or experimental study. It evaluates an AI-assisted workflow using published research and student feedback. Any effectiveness claim must be based on feedback actually collected during the project.

5. Selected Open-Access Research Papers

The three papers below focus on AI in higher education, so the project remains within a relevant IT and AI topic. Open each source and read the full article before treating the summary as final.

Paper 1 — Artificial intelligence in education: A systematic literature review (Wang et al., 2024)

Source: Expert Systems with Applications, Volume 252, article 124167. DOI: 10.1016/j.eswa.2024.124167. Open-access article: https://www.sciencedirect.com/science/article/pii/S0957417424010339

Research question in simple words: What types of AI are being used in education, what topics are researchers studying, and what methods do they use?

Method: The authors conducted a systematic literature review, organizing previous studies to identify application categories, common research topics, findings, and research-design patterns.

Beginner-level explanation: Instead of testing one new AI application, the researchers reviewed existing research to build a wider picture of AI in education. This kind of review helps readers see repeated themes and gaps across many studies.

Key takeaway: AI in education is a broad field with different application areas and research approaches. A reader should not assume every AI tool or every classroom use has the same effect; claims depend on the specific tool, context, and evidence.

Limitations to check in the original: Review conclusions depend on which databases, search terms, dates, and inclusion criteria were used. A review may not cover every relevant study, and findings can become outdated as tools change.

Paper 2 — Systematic literature review on opportunities, challenges, and future research recommendations of artificial intelligence in education (Chiu et al., 2023)

Source: Computers and Education: Artificial Intelligence, Volume 4, article 100118. DOI: 10.1016/j.caeai.2022.100118. Open-access article: https://doi.org/10.1016/J.CAEAI.2022.100118

Research question in simple words: How has AI been used in learning, teaching, assessment, and education administration, and what benefits and problems have researchers identified?

Method: The authors reviewed research from 2012–2021 and used structured coding and content analysis to group AI roles, learning outcomes, and challenges.

Beginner-level explanation: The authors organized past research into categories instead of focusing on one product. This makes it easier to understand where AI may help and what issues schools should plan for.

Key takeaway: AI may support different educational tasks, but responsible implementation needs to consider challenges and the quality of evidence. A benefit reported in one setting should not automatically be generalized to every course or student.

Limitations to check in the original: The review's search window ends in 2021, so it cannot fully represent later developments in generative AI. As with all reviews, its results depend on the selected studies and classification process.

Paper 3 — Design and assessment of AI-based learning tools in higher education: a systematic review (Luo et al., 2025)

Source: International Journal of Educational Technology in Higher Education, Volume 22, article 42. Published 11 July 2025. DOI: 10.1186/s41239-025-00540-2. Open-access article: https://link.springer.com/article/10.1186/s41239-025-00540-2

Research question in simple words: How are AI learning tools designed for college students, and what effects do they have on learning?

Method: The authors followed a systematic review approach using PRISMA guidelines. They reviewed 63 peer-reviewed articles published between January 2014 and April 2024.

Key findings reported by the paper: The review identified common AI roles such as assessment and evaluation, personalized feedback and recommendations, and intelligent tutoring. The authors report that tools generally supported knowledge acquisition and affective outcomes, while results for higher-level cognitive processes and skills varied.

Beginner-level explanation: AI tools can quiz students, provide feedback, recommend what to study next, and act like a tutor. But doing better on a quiz does not automatically mean a student has improved every skill, such as deep reasoning or independent problem-solving.

Limitations to check in the original: The review includes studies only up to April 2024. Studies use different tools, subjects, and evaluation methods, so results may not be directly comparable. Some tools may work well in one context but not another.

6. Comparison of the three papers

Paper	Main focus	Method	What a student learns
Wang et al. (2024)	Broad overview of AI in education	Systematic literature review	AI education research contains multiple themes and research approaches.
Chiu et al. (2023)	Opportunities, challenges, and future research	Systematic review and content analysis	AI can serve several educational roles, but implementation has challenges.
Luo et al. (2025)	Design and learning outcomes of AI tools in higher education	Systematic review of 63 articles	Feedback, assessment, and tutoring are common; outcomes vary by skill and context.

Synthesis: All three papers indicate that AI has multiple potential roles in education, but AI use should be evaluated carefully. The strongest project conclusion is not that AI always improves learning; it is that AI may help explain and support learning when used with suitable evidence, human judgment, and transparent checks.

7. Proposed project methodology

1. Select the three open-access papers listed in Section 5 and save the full PDFs or article pages.
2. Read each abstract, introduction, methods, results, and limitations section.
3. Upload one paper at a time to NotebookLM or another approved AI tool.
4. Use the prompt library in Section 8 to request summaries, definitions, methods, results, and FAQs.
5. Compare each AI statement with the source paper and record any error, omission, or unsupported claim.
6. Create a one-page visual summary for each paper using Canva or another design tool.
7. Prepare a five-slide presentation about one paper.
8. Ask classmates to read the original abstract and AI summary, then complete the feedback form.
9. Calculate feedback scores only after collecting real responses.
10. Write the final conclusion based on the evidence and feedback actually collected.

8. Prompt library

Use these prompts with one paper at a time. Keep a copy of the exact prompt and the tool/version/date used.

Prompt 1 — Beginner summary

"Explain this research paper for an undergraduate IT student in simple English. Use only information supported by the paper. Include the research problem, objectives, method, main findings, limitations, and conclusion. If a detail is not stated, write 'not clearly reported' rather than guessing. Cite the section or page supporting each important claim."

Prompt 2 — Explain methodology

"Explain the methodology step by step. Define technical terms, identify the data or studies reviewed, explain how the authors analyzed the evidence, and distinguish what the authors did from what they recommend. Do not invent sample sizes, metrics, or results."

Prompt 3 — FAQs

"Create five beginner-level questions and answers about this paper. Include at least one question about the method, one about findings, and one about limitations. Keep every answer grounded in the paper and mention where I can verify it."

Prompt 4 — Verification

"Check the following summary against the provided paper. For every sentence, classify it as supported, partly supported, unsupported, or contradicted. Give the relevant source section/page where possible. Do not silently rewrite errors; list them separately."

Prompt 5 — Visual summary content

"Create concise text for a one-page academic infographic: title, problem, objective, method, key findings, limitations, and takeaway. Do not add any fact that is not in the paper. Keep each section to one or two short bullet points."

9. AI-assisted workflow

Stage	Tool / action	Output	Human check
1. Collect	Open-access journal pages / PDF	Three source papers	Confirm article title, authors, year, DOI, and access.
2. Explain	NotebookLM / ChatGPT / Gemini	Summary, glossary, FAQs	Compare each claim to the paper.
3. Visualize	Canva	Three one-page summaries	Check wording, figures, and attribution.
4. Present	Gamma or PowerPoint	Five-slide deck	Remove unsupported claims and add references.
5. Evaluate	Feedback form and spreadsheet	Understanding scores	Use real responses; report sample size.
6. Document	Prompt log and error log	Verification report	Keep date, tool, prompt, error, and correction.

10. One-page visual summary drafts

Visual summary A — AI in Education: A Systematic Literature Review

Problem	Method	Key message	Limit / caution
Research on AI in education is spread across different topics.	Review of existing literature and research designs.	A systematic review organizes the field into themes and shows how researchers study AI in education.	Coverage depends on search strategy, time period, and included studies.

Visual summary B — Opportunities and Challenges of AI in Education

Problem	Method	Key message	Limit / caution
Educators need a clear view of AI's roles and risks.	Review of studies from 2012–2021 using coding and content analysis.	AI can support several educational tasks, but benefits and challenges should be assessed in context.	The review predates many recent generative-AI developments.

Visual summary C — Design and Assessment of AI Learning Tools

Problem	Method	Key message	Limit / caution
Design and learning effects of AI tools are not always clear.	Systematic review of 63 peer-reviewed articles.	AI tools commonly provide assessment, personalized feedback, recommendations, and tutoring; learning outcomes vary.	Evidence varies across subjects, designs, and outcomes.

11. Five-slide presentation outline

Suggested title: “Can AI Make Research Papers Easier to Understand?” Use Paper 3 as the main example and briefly compare it with Papers 1 and 2.

Slide	Content	Suggested speaker note
1. Introduction	Project title, problem, aim, and selected paper.	Research papers can be difficult for beginners; this project tests an AI-assisted explanation workflow.
2. Research question	How are AI learning tools designed, and what learning outcomes do they affect?	The selected review examines tools used in higher education and evaluates reported outcomes.
3. Method	Systematic review; 63 peer-reviewed articles; period January 2014–April 2024.	The authors used a structured review process rather than testing one AI tool on their own.
4. Findings	Assessment/evaluation, personalized feedback/recommendations, intelligent tutoring; mixed results across outcomes.	AI may support knowledge acquisition, but higher-level thinking and skill outcomes vary.
5. Conclusion and limitations	AI can support learning; verify outputs; results depend on tool, context, and evidence.	AI should assist learning rather than replace reading, critical thinking, or teacher guidance.

12. AI verification and inaccuracy log

Fill this table only after checking the actual AI-generated outputs. The example below is a hypothetical example showing how to record an error; it is not a claim that the tool made this exact mistake.

Date / tool	AI statement	Verification result	Correction / evidence
To be completed	To be copied exactly from AI	Supported / partly supported / unsupported / contradicted	Corrected wording and page/section in source
Example only	“The review proves every AI tool improves all student skills.”	Unsupported overgeneralization	The reviewed paper reports generally positive findings for some outcomes, but results for cognitive processes and skills vary. Check the Results section of Luo et al. (2025).

Verification checklist

- Does every number match the source?
- Are the paper’s method and sample described accurately?
- Does the summary distinguish findings from recommendations?
- Are limitations and uncertainty included?
- Are quotations, figures, and borrowed ideas attributed?
- Are citations real and linked to the correct source?
- Did I mark claims that could not be verified?

13. Classmate feedback questionnaire

Give participants the same short original-paper excerpt and the AI summary. Ask them to rate each statement from 1 (strongly disagree) to 5 (strongly agree). Obtain any required teacher permission and do not collect unnecessary personal information.

Statement	Rating (1–5)
The main problem of the paper is clear.	_____
I understand the method used by the researchers.	_____
I can explain the main finding in my own words.	_____
The summary is easier to understand than the original excerpt.	_____
The summary includes important limitations.	_____
I would use this type of explanation as a study aid.	_____

Feedback recording table

Participant code (no names)	Before score	After score	Notes
P01	_____	_____	_____
P02	_____	_____	_____
P03	_____	_____	_____
P04	_____	_____	_____
P05	_____	_____	_____

Calculate an average only after you collect actual responses. Average score = sum of scores ÷ number of valid responses. Do not invent feedback or claim improvement without data.

14. Before-and-after analysis

Compare a paragraph from the original paper with the AI explanation. Keep the source excerpt short and properly attributed. Record the word count, readability, key facts preserved, and any errors found.

Measure	Original paper excerpt	AI explanation	How to evaluate
Word count	_____	_____	Was the explanation shorter without losing essential meaning?
Technical terms	_____	_____	Were terms explained correctly?
Key facts preserved	_____	_____	Check research question, method, results, and limitations.
Unsupported claims	Not applicable	_____	Count and document claims not supported by the source.
Student understanding	Before quiz: _____	After quiz: _____	Compare actual scores; report the number of participants.

15. Responsible AI and ethics

- **Accuracy:** AI summaries must be checked against original sources. Never present an unverified generated claim as a research finding.
- **Privacy:** Use public research papers; do not upload personal student records or confidential information.
- **Copyright:** Respect article licenses and publisher terms. Summarize in your own words, cite sources, and do not redistribute a paper unless its license permits it.
- **Honesty:** Disclose which AI tools were used and what they helped produce, following college rules.
- **Bias and limitations:** A summary can omit context or overstate a conclusion. Human review is essential.
- **Academic integrity:** Understand the papers yourself and be ready to explain the work during the demonstration.

16. Tools and skills used

Tool	Planned use	Skill demonstrated
NotebookLM	Work with uploaded papers and ask source-grounded questions.	Research and source-based prompting
ChatGPT	Simplify methods, draft FAQs, and check clarity.	Prompt engineering and content creation
Gemini	Produce a second explanation for comparison.	Cross-checking and verification
Canva	Design three visual summaries.	Visual communication
Gamma / PowerPoint	Create a five-slide presentation.	Presentation design
Spreadsheet	Calculate real feedback averages and maintain logs.	Data organization and analysis

17. Expected outcomes

- A repeatable workflow for explaining research papers at beginner level.
- Three readable summaries and three visual summary drafts.
- A five-slide presentation outline.
- A prompt library and a verification log template.

- A feedback process that can be used to assess understanding.
- A responsible-AI discussion that recognizes both benefits and limitations.

18. Conclusion

This project presents a structured way to use AI as a research-learning assistant. The selected literature shows that AI has several roles in higher education, including assessment, personalized feedback, recommendations, and tutoring. However, the quality and impact of AI tools vary across learning contexts and outcomes. The proposed workflow therefore combines AI-generated explanations with direct reading, source verification, visual communication, and student feedback. The final conclusion about usefulness should be written after the student completes the verification process and collects real feedback.

19. Limitations of this project

- The initial report summarizes published papers; it does not independently reproduce their review methods.
- AI tools and free-tier features can change over time.
- Feedback from a small class group may not represent all undergraduate students.
- Perceived clarity is not the same as long-term learning or research competence.
- Results should not be generalized beyond the selected papers and the actual evaluation sample.

20. Final demonstration plan

1. Introduce the problem and the project objective.
2. Open one original open-access paper and identify its research question, method, findings, and limitations.
3. Show the prompt used in NotebookLM or another AI tool.
4. Display the AI summary and explain how it was verified against the paper.
5. Show the one-page visual summary and five-slide presentation.
6. Present the completed inaccuracy log and prompt library.
7. Show the real feedback results, including participant count and average scores, if collected.
8. End with limitations, ethics, and lessons learned.

21. Quick viva questions and answers

Q1. What is the aim of your project?
A: To use AI tools to explain complex research papers in simple language and verify the explanations against the original sources.

Q2. Why did you select open-access papers?
A: They can be read and checked by students without a subscription, subject to their license terms.

Q3. What is a systematic literature review?
A: It is a structured method for finding, selecting, and analyzing previous studies using stated procedures.

Q4. What is an AI hallucination?
A: A generated statement that sounds plausible but is false or unsupported by the available evidence.

Q5. How do you verify an AI summary?
A: Compare each important claim with the original paper, record the supporting section or page, and correct unsupported statements.

Q6. What are the limitations of your project?
A: The findings depend on three selected papers, the AI tools used, and the size and quality of the feedback sample.

Q7. Why use more than one AI tool?
A: Different tools may explain information differently. Comparing them can reveal inconsistencies, but agreement between tools does not replace checking the original source.

Q8. What is the main conclusion?
A: AI can help make research more accessible, but accurate understanding requires source checking and independent student thinking.

22. References

[1] Wang, S., Wang, F., Zhu, Z., Wang, J., Tran, T., & Du, Z. (2024). Artificial intelligence in education: A systematic literature review. *Expert Systems with Applications, 252*, 124167. https://doi.org/10.1016/j.eswa.2024.124167

[2] Chiu, T. K. F., Xia, Q., Zhou, X., Chai, C. S., & Cheng, M. (2023). Systematic literature review on opportunities, challenges, and future research recommendations of artificial intelligence in education. *Computers and Education: Artificial Intelligence, 4*, 100118. https://doi.org/10.1016/j.caeai.2022.100118

[3] Luo, J., Zheng, C., Yin, J., et al. (2025). Design and assessment of AI-based learning tools in higher education: A systematic review. *International Journal of Educational Technology in Higher Education*, 22, Article 42. https://doi.org/10.1186/s41239-025-00540-2

Reference-use note: Read each complete source before final submission. Bibliographic details and summaries should be checked against the publisher page. Follow the source's license and your institution's citation rules.

Appendix A — Final submission checklist

- Student and college details completed on cover page.
- Full text of all three papers read and saved for reference.

- Three AI summaries checked against original papers.
- Three visual summaries designed and checked.
- Five-slide presentation finalized.
- Prompt library updated with the prompts actually used.
- Verification log completed with real examples and source locations.
- Feedback data collected ethically and calculated accurately.
- Before-and-after analysis completed.
- AI tools and assistance disclosed according to college rules.

End of report

AI for Complex Problem-Solving

problem statement

Artificial Intelligence can help people solve complex problems, but its role in human thinking, teamwork, emotions, and self-monitoring is not fully understood.

objectives

To review existing research and identify how AI supports human complex problem-solving, including cognitive, emotional, and social processes.

Method used

The study used a scoping literature review of Scopus database publications from 2017 to 2021, selecting 38 studies for coding and analysis.

Result

AI support was studied most often for cognitive tasks, such as analysing information and making decisions .The four dimensions—cognitive, metacognitive, social, and affective

Conclusion

AI can help humans solve complex problems, but further research is needed to improve self-monitoring, teamwork, and emotional support.

Opportunities of Artificial Intelligence for Supporting Complex Problem-Solving

Method used

The study analyzed 38 out of 2,607 Scopus studies (2017–2021) to examine four dimensions of human–AI collaboration.

Objectives

To understand how AI supports humans in solving complex problems through cognitive, social , etc.

Result

AI research focuses mainly on thinking support, while social and emotional support needs improvement.



problem statement

Complex problems are difficult to solve, and AI can help, but its role in teamwork, emotions, and thinking needs more research.

Conclusion

AI supports problem-solving and teamwork but should complement, not replace, human judgment.

A Non-Price-Based Tender Evaluation Methodology Considering Tenderers' Self-Confidence with Partial Value Functions and Time Decay Analysis

Problem statement

Choosing contractors by lowest price alone can cause delays, extra costs, poor quality, and disputes.

Objective

To develop a tender evaluation model that considers both price and non-price factors to identify the most suitable construction contractor.

Criteria selection

Literature review and expert ratings were used to identify criteria and calculate their importance using RII.

Model testing

scoring model with partial value functions and time-decay correction was tested on nine contractors.

Result

Tender No. 8 was the best choice, showing that price, criteria weights, safety, quality, and past performance affect contractor selection.

conclusion

The model improves contractor selection using price, quality, and past performance, but needs reliable data and further real-world testing.